

Problema 2.12

Presupunem ca $A \in \mathbb{R}^{n \times n}$ o matrice nesingulara si $b \in \mathbb{R}_*^n$. Presupunem ca $Ax=b$ si $A(x + \delta x) = b + \delta b$

,teoria numarului de conditinoare afirma ca:

$$\frac{\|\delta x\|}{\|x\|} \leq \text{cond}(A) \frac{\|\delta b\|}{\|b\|}$$

Considerand vectorii proprii ai matricei $A^T A$, aratati cum se pot gasi vectorii b si δb pentru care egalitatea are loc in norma $\|\cdot\|_2$.

Se stie ca

- 1) $\|A\|_2 = \max_{v \in \mathbb{R}_*^n} \frac{\|Av\|_2}{\|v\|_2} = \sqrt{\lambda_n}$, unde $\lambda_1 \leq \dots \leq \lambda_n$, λ_i $i \in \overline{1,n}$ valori proprii ale $A^T A$
- 2) $\text{cond}(A)_2 = \|A\|_2 \|A^{-1}\|_2$

Calculam $\|A^{-1}\|_2$.

$$\|A^{-1}\|_2 = \max_{v \in \mathbb{R}_*^n} \frac{\|A^{-1}v\|_2}{\|v\|_2} = \sqrt{\mu_n}, \text{ unde } \mu_1 \leq \dots \leq \mu_n, \mu_i \text{ } i \in \overline{1,n} \text{ valori proprii ale } A^T A^{-1}$$

Stiim ca daca

λ valoare proprie a vectorului X atunci $\frac{1}{\lambda}$ este valoare propriea lui X^{-1} . Rezulta ca $\mu_1 = \frac{1}{\lambda_n} \leq$

$$\dots \leq \mu_n = \frac{1}{\lambda_1} \text{ si deci } \|A^{-1}\|_2 = \frac{1}{\sqrt{\lambda_1}}, \text{ deci } \text{cond}(A)_2 = \sqrt{\frac{\lambda_n}{\lambda_1}}.$$

Se alege $x = \text{vectorul propriu al lui } \lambda_n$ si $\delta x = \text{vectorul propriu al lui } \lambda_1$.

Ecuatiile devin:

$$\lambda_n x + \lambda_1 \delta x = b + \delta b \text{ si}$$

$$\frac{\|\delta x\|}{\|x\|} \leq \sqrt{\frac{\lambda_n}{\lambda_1}} \frac{\|\delta b\|}{\|b\|}$$